The Impact of RFID Technology on Supply Chain Management: A Literature Review

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Abstract

Utilizing RFID technology offers many benefits to the supply chain. Strategic implementation of RFID tags, readers, frequencies, and levels of tagging can utilize its unique ability to automatically scan and track without being in the line-of-sight of the scanner to reduce inventory errors, labor reduction, and create an overall more efficient process. With a wide range of RFID systems and applications, this has been a popular topic of research to optimize supply chain management. 10 papers with a focus on the impact of RFID technologies on supply chain management were reviewed and analyzed. A variety of papers with a mixture of analytical, simulation, and case study approaches were chosen. Finally, areas of future studies and conclusions are discussed.

Keywords
RFID Technology, Supply Chain Management, ROI, RFID Adoption

1. Introduction
RFID stands for Radio-Frequency Identification and is a small electronic device that consists of a small chip and an antenna and uses electromagnetic fields to automatically identify and track tags attached to objects. Much like a barcode or magnetic strip on a bankcard, RFID tags provide products with a unique identification. Although RFID and barcode systems are comparable, there are many advantages and benefits RFID has over the traditional system. The major benefit behind RFID implementation is that each tag will work within a few feet of the sensor. For example, RFID sensors would be able to simultaneously scan and log pertinent information on all incoming products from a truck shipment, saving significant time and reducing operator error. The applications to which RFID technology may be applied to for increased efficiency are endless. An additional example is an innovative implementation of RFID tags in a grocery store to eliminate the checkout line completely. This concept is so valuable that there are currently patent filings describing a system that uses technology including RFID to detect when a shopper takes an item from the shelf, and then syncs the data to a handheld device [1]. With such a wide range of opportunity for RFID to improve efficiency along a process, it has become a heavily researched topic. In particular, a lot of research has been carried out relating to RFID system implementation in supply chain management. The study of supply chains covers every process involved in the flow of products from manufacturing to the customer's hands. With supply chain management covering marketing decisions, manufacturing, distribution, and transportation, a technology that can assist in the efficiency of this huge task would be incredibly beneficial. This is why RFID technology has grown into a revolutionary aspect of supply chain management. It has the capability to significantly reduce inventory errors and increase reliability of the entire chain. RFID offers an overall faster authentication of produced goods, increased information gathered with the help of RFID technology, and storage of said information which can be used for further efficiency planning [2]. Current research is in the field of inventory inaccuracy, the bullwhip effect, and replenishment policies. In efforts to develop and analyze the effect RFID technology has on supply chain management, models are used in the form of analytical, simulations, and case studies. In addition to modeling the optimal supply chain, Return-on-investment (ROI) analyses are often focused on, as the ultimate question is always 'is it *really* worth it?'. Companies with the opportunity to implement RFID technologies into their supply chain would find it beneficial to review current research regarding the vast benefits and overall potential savings that come along with such a change.

2. Possible applications and benefits of RFID Systems in supply chain management

Texts related to the numerous benefits of implementing RFID systems into a supply chain were collected and reviewed. It is noted that current studies of RFID in supply chains focus on inventory management, logistics and transportation, assembly and manufacturing, asset tracking and object location, environment sensors, etc [3]. Although, there are countless industries that may benefit from these applications as well, such as retail, healthcare, textile, automotive, and luxury good industries. Benefits mentioned may be realized as beneficial regardless of the industry.

2.1 RFID: An enabler of supply chain operations by Mohsen Attaran

Attaran does an excellent job of providing a conceptual discussion of the evolution of RFID while also addressing implementation challenges and reviewing success factors. An interesting implication is made that the technology is in its infancy, which has yet to tap into its potential for supply chain collaboration. A few examples of areas in which a variety of industries may benefit from RFID technology is discussed at length. For example, in shipping and distribution, this technology enables suppliers to accurately determine the location of a pallet, track its journey through the chain, and to make instantaneous routing decision [4]. Alternatively, in the retail industry, RFID tags appear to continuously have an advantage over barcoding. Attaran writes that Wal-Mart can save $6.7 billion in labor costs alone from RFID implementation as well as continually gather information for reordering, stocking, and keeping track of purchases [4]. The manufacturing sector finds value in RFID to track the usage, availability, location, and maintenance of material handling equipment. For example, Proctor & Gamble (P&G) has begun to track where every item is in the manufacturing process and supply chain and expects cost saving up to $1 billion in working capital and $200 million in inventory carrying costs and will ultimately return their investment as well as help P&G's bottom line [4]. A real world application of RFID technology is the use of wristbands in the U.S Navy to improve quality and reliability through easier patient identification and automatic status updates. A more crucial use would be the tagging of medications and prescription doses that could ensure the correct medicine is given in the proper dose at the correct time. In the pharmaceutical industry, Purdu Pharma, the manufacturer of OXYContin uses RFID tagging to track shipment of valuable medication. Even the government agencies are able to find use for tagging on supply containers, which detect shock and temperature variance, and also for weapon management for use relaying materials, capabilities, and mission details [5]. The gaming industry has found clever ways to use RFID tagging to their advantage; Wynn Las Vegas Casino uses radio tags on betting chips to deter counterfeiting, card-counting, and other illegal behavior. In addition, the dealers are able to easily take a quick inventory of chips and monitor movement of chips and bets. RFID will more accurately and efficiently rate players and utilize this data to target desired customers [4]. The argument made by Attaran that RFID has a lot of untapped potential and can be used for an astounding array of applications is a compelling one; through multiple examples of real world application of RFID adoption and how it benefited various industries, valuable lessons were learnt regarding supply chain management and how RFID is advantageous in many cases.

2.2 Strategic value of RFID in supply chain management by May Tajima

This paper aims to provide insight into the strategic value of radio frequency identification (RFID). Tajima makes a valid point that there is a large gap in available research between the actual and potential values of this technology and builds a theory on how RFID used in supply chain management may create and sustain a competitive advantage [6]. A theory of organizational learning and the theory of first-mover advantages were the key components in four propositions developed to demonstrate this potential value. Beginning with a striking parallel in which Tajima points out, "following the RCA-Kroger pilot program and standardization of the Universal Product Code (UPC), fewer than 1% of grocery stores in the United States were using bar codes in the late 1970s. The usage then jumped to more than 30% after Wal-Mart endorsed the use of barcodes in 1984"[6], she addresses the fact that Wal-Mart began requesting that its top suppliers implement the use of RFID systems. This parallel drawn out by the author, speaks to the revolutionary potential of this technology. However, it comes as a surprise that even with large promises such as shrinkage reduction, material handling efficiency, and improved asset management, RFID has been slow to adopt across the wide variety of supply chain management application; Tajima argues that this is due to the aforementioned gap in the reality versus potential perceptions of RFID. With many researchers trying to fill this void with quantifiable data, real-life setting case studies, and analytical models, the author approaches the issue from a different standpoint; "Does the use of RFID create a competitive advantage for a firm, that is, an ability to perform better than the firm's competitors? Is the competitive advantage sustainable?" [6]. Tajima builds her theory in four distinct parts. First, the potential benefits of RFID are laid out and discussed. Second, from the benefits previously

mentioned, a conceptual framework is developed. Third, patterns observed from the framework (process automation, closed-loop tracking and supply chain visibility) are used to develop two propositions regarding how RFID may create a competitive advantage. Finally, two propositions are discussed regarding how RFID implementation may achieve a competitive advantage that is not only sustainable, but profitable as well. Among some of the potential benefits gathered from an extensive literature review are reduced shrinkage, reduced material handling, increased data accuracy, faster exception management, improved information sharing, production tracking, quality control, and supply and production continuity [6]. Further, from the fundamental characteristics of RFID systems and how their patterns of use might result in various benefits, Fig. 1 shown below was developed.

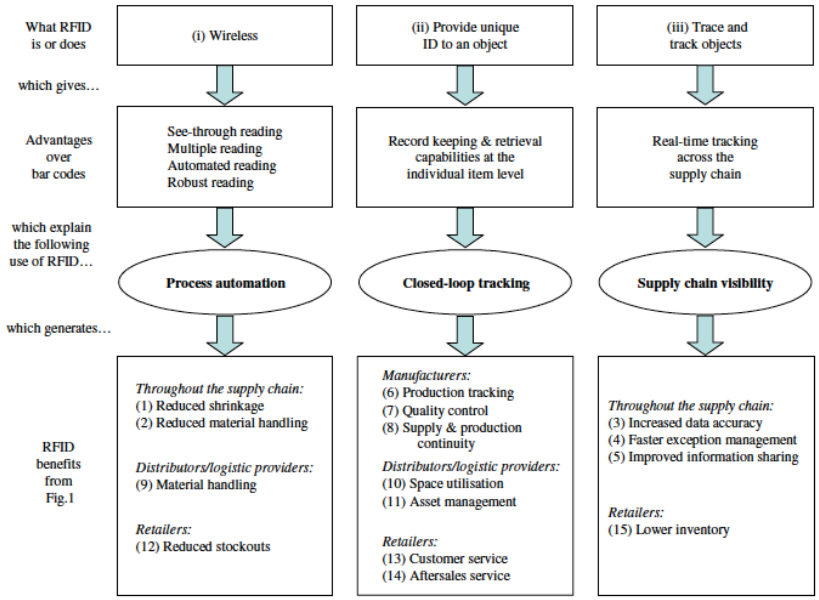

Figure 1. Alternative Framework for RFID Benefits [6]

From this framework, an exploitation-exploration perspective was used to develop two propositions. The first proposition provides that "the use of RFID for exploitation improves the firm's competitive advantage through increased supply chain efficiencies" [6]. The author's second proposition is that "the use of RFID for exploration improves the firm's competitive advantage through increased innovation capability" [6]. The IT theory of sustaining the first mover advantages by remaining ahead of the learning curve was used to develop the next two propositions;

Proposition 3 (P3). *The long-term use of RFID for exploitation improves the firm's short-term competitive advantage through ''learning to adjust''.*

Proposition 4 (P4). *The long-term use of RFID for exploration improves the firm's long-term competitive advantage through ''learning to transform''.*

Together, the four propositions dive deeper into the strategic value of RFID implementation and provide multiple ways to achieve competitiveness and sustainability. The identifiable limit to this research area is the lack of extensive empirical data that would be necessary to support the validity of the propositions. However, even without empirical data, compelling discussions of the innovative potential of RFID implementation, as well as four insightful propositions were presented. The author concludes with a suggestion that RFID may be analyzed using alternative methods such as using RFID in the transaction between trading partners to reduce transaction costs[6].

2.3 The impacts of Radio Frequency Identification (RFID) technology on supply chain costs by Alp Ustundag

This research paper uses a simulation model to predict the benefits of an integrated RFID supply chain in terms of cost factors at the echelon level. With increases in efficiency, accuracy, visibility, and security level, the study looks into the improvements in manufacturing and retail operations that could be made from RFID implementation. To

evaluate the impact of RFID five fundamental processes were considered: receiving, put away, inventory counting, packing, and shipping [7]. The supply chain for this research paper was a pull-based system consisting of manufacturer, distributor, and retailer out of a retail textile supply chain in Turkey. With a plan to integrate an RFID system in its warehouses, this was a great opportunity for a simulation model to ensure it an economically sound decision. Figure 2 shown below depicts the process flow of the simulation model with the expected benefit areas along the output.

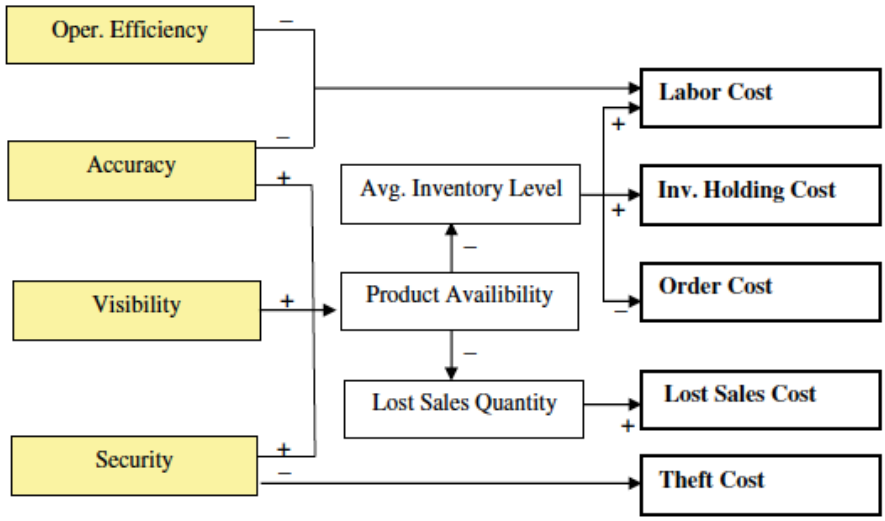

Figure 2. Supply chain cost factors [7]

The model is able to calculate the value change in cost of labor, inventory holding, orders, lost sales, and theft for an integrated RFID system. Each total process cost is able to be managed an analyzed using the model and overall benefits are determined by total cost factors for each member of the supply chain. Although many parameters involved in this simulation are fixed, there are a few which were estimated. For example, end-customer demand was independently and identically normally distributed, with a daily average of 1000 items [7] . With three different groups of hypotheses, multiple ANOVA analyses, and calculation of overall process costs and possible error, the conclusion was made that increasing lead time decreases the total cost savings to the retailer in the integrated supply chain. Lead time did not appear to have any meaningful impact on total cost savings and rising inventory levels increased labor cost [7]. With the third set of hypotheses being supported, the simulation concluded that the rising level of demand uncertainty decreases the total cost savings at each level of the integrated RFID supply chain [7].

3. Approaches used to evaluate implementation of RFID

Determining a suitable approach to evaluate an RFID implementation plan requires a conceptualized look at the entire system at hand. With so many possibilities and opportunities for RFID implementation, there is not a one size fits all approach. Whether it is a simulation model used to predict if the integration of this technology would be valuable enough to the supply chain, or an analytical approach to an already existing system's efficiency, evaluation and research of benefits and outcomes of implementation of RFID is a very opportune research topic.

3.1 A simulation approach to demonstrate integration of RFID

Simulation approaches aid in providing a framework to determine conditions in which RFID implementation would be beneficial to that specific process. An article in which a four-echelon supply chain is operated under different levels of collaboration was analyzed using a comprehensive simulation model was published in the European Journal of Operational Research. This article does an excellent job illustrating that the integration of RFID technology within a supply chain provides many benefits, especially when collaboration between the participants is more involved [8]. Additionally, it was found that longer lead times and certainty of customer demand plays an important role in the benefits of implementation. Sari utilized an analysis tool called Monte Carlo simulation by using Crystal Ball, a commercial spreadsheet-based risk analysis and forecasting program published by Decisioneering [8]. Consisting of four-echelons: a factory, a warehouse, a distributor, and a retailer, the simulation

model generates demand, determines available manufacturing capacity, forecasts demand, plans production and inventory, and fulfills/reports orders. Fig. 3 below shows the logic behind the simulation and summarizes activities performed throughout along the supply chain.

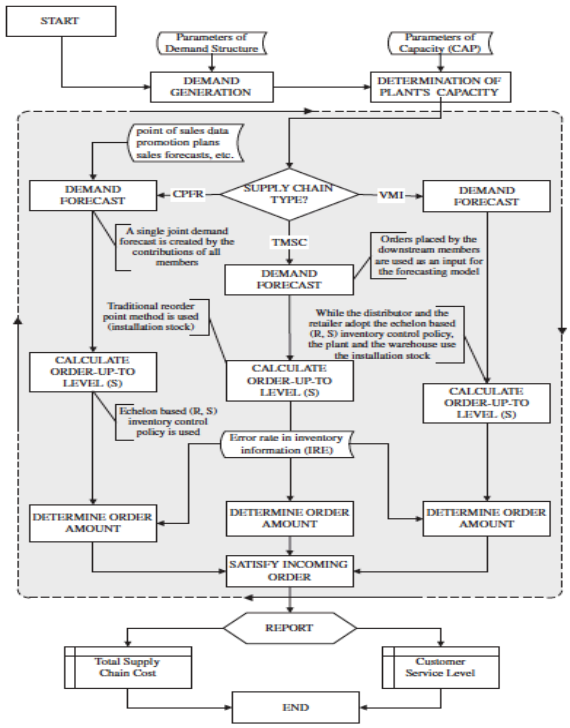

Figure 3. Flowchart of simulation model [8]

Verification of the model was performed through manual calculations as well as testing and tracing each step. Additionally, the random demand variables were plotted on a scatter diagram and it was confirmed that the demand structures were generated as intended. From the simulation, results were found which implicated that benefits were optimized when operators were participative, uncertainty in customer demand is lower, and lead times along the supply chain are longer. Capacity levels were found to not have a significant effect on performance. Sari suggests future studies include a more complicated supply chain structure such that different types of inventory and production policies and replenishment decisions are considered.

3.2 An analytical model of RFID adoption in a two-echelon supply chain

Szmerekovsky's paper focuses on the possibility for the manufacturer and the retailer to derive economic benefits from item-level RFID [9]. The first case evaluated consists of the manufacturer setting the wholesale price for the product and the retailer setting the retail price as well as allocate shelf-space. The second case considered involves a powerful retailer who settles product shelf-space prices and the manufacturer is simply a renter of space from the retailer. The model used to analysis of RFID adoption takes into account numerous variables such as revenue, satisfied demand, uncensored consumer demand, as well as various costs along the supply chain. The analytical model was used to evaluate the impact of RFID technology in affect of manufacturing costs, shelf management costs, and ability to satisfy customer demand. In addition, the demand model and shelf-space management cost model and supply chain model are evaluated, as there is an interest in investigating how decentralization of the supply chain affect the decisions about RFID adoption. Finally, an analysis of the three supply chain scenarios with respect to adoption of item-level RFID was performed and conclusions were made. An integrated supply chain, a wholesale price contract, and a shelf-space renting contract successfully covers a wide range of conditions to confidently draw conclusions from. From these analytical models, two main results were reached; first, if the manufacturer is able to pass on the cost of RFID tagging to the retailer through a higher wholesale price, then the interests of both retailer and manufacturer in terms of profitability and second, when there is significant competition

for shelf-space and the manufacturer is unable to pass some of the costs onto the retailer, then the retailer may benefit from the tags while the manufacturer might not [9]. Suggestions from the results include that each member's power in the supply chain plays a role in RFID adoption.

4. Conclusion

Showcasing several journal articles relating to the integration of RFID systems to assist in supply chain management, this paper aimed to summarize the current research available which proves RFID to be a valuable tool for manufacturing facilities to retail storefronts. While there are countless benefits and application opportunities for RFID implementation, it would be beneficial to first consider a modeling approach, which would optimize all variables such that the full potential of RFID is utilized. Future research suggestions include more complex simulations, analytical models with decreased uncertainty within variables, and further research between the actual and potential values of RFID technology implementation along a supply chain. Further, theories must be developed related to how RFID used in supply chain management may create and sustain a competitive advantage

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